

Analysis Qualifying Exam, January 2026, Part 1

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do three of four.

1. Do two of three.

- (i) Give an example of a Lebesgue measurable set $E \subseteq [0, 1]$ that is closed, has positive measure, but contains no non-trivial interval or explain why no such example exists.
- (ii) Is the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = \frac{1}{1+x^2}$ the Hardy–Littlewood maximal function of an $\mathcal{L}^1(\mathbb{R})$ function f ?
- (iii) Does there exist a measure space $(X, \mathcal{M}, \mu)$ and a nested decreasing sequence $(E_n)_{n=1}^\infty$ of measurable sets such that

$$\lim_{n \rightarrow \infty} \mu(E_n) \neq \mu\left(\bigcap_{n=1}^\infty E_n\right)?$$

2. Suppose $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$. Show, if (a)–(c), then $F : [0, 1] \rightarrow \mathbb{R}$ defined by

$$F(x) = \int_{[0,1]} f(x, y) d\lambda(y)$$

is continuous. Here λ denotes Lebesgue measure on $\mathbb{R}$.

- (a) for each $x \in [0, 1]$ the function $g_x(y) = f(x, y)$ is continuous;
- (b) for each $y \in [0, 1]$ the function $h_y(x) = f(x, y)$ is continuous; and
- (c) f is bounded,

3. Let $(X, \mathcal{M})$ denote a measurable space and let $P(X)$ denote the power set of X . Prove, if $\mathcal{M} \neq P(X)$ and $\{x\} \in \mathcal{M}$ for each $x \in X$, then there exists an $A \notin \mathcal{M} \otimes \mathcal{M}$ such that $[A]_x, [A]^y \in \mathcal{M}$ for all $x \in X$. Suggestion: First show $\delta : X \rightarrow X \times X$ defined by $\delta(x) = (x, x)$ is measurable from $(X, \mathcal{M})$ to $(X \times X, \mathcal{M} \otimes \mathcal{M})$.

4. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $f, g : X \rightarrow [0, \infty)$ are measurable. Show

- (a) $\mu_f : \mathcal{M} \rightarrow [0, \infty]$ defined by $\mu_f(A) = \int \chi_A f d\mu = \int_A f d\mu$ is a measure; and
- (b) $\int g d\mu_f = \int gf d\mu$.